Why Our Knowledge Looks Like This: Representation, Path Dependence, and a Question from High School

In high school, I started asking a question that no one around me could really answer: why do we have to learn this? I did not only mean subjects I was not interested in. I meant something larger. Why does our knowledge look the way it does? Why are mathematical formulas written in this particular way, and why did our civilization develop these sciences and social structures rather than others? If the earliest civilizations had taken a different path, would we now be learning completely different things? I even imagined parallel universes in which history had gone another way. Even when a theorem is proved, the truth it describes was already there before anyone proved it. So why does that truth appear to us in this particular form? At the time, these questions felt like a trap. I still had to study for exams, and asking why only made studying harder.

Studying cognitive science has given me words for what I was asking. Theorems, formulas, and number systems are representations: forms in which knowledge is written down. Once created, they are used for computation, to solve problems, build tools, and produce new representations, which are used in turn. Each round builds on the one before. What I was really asking was whether this process had to arrive where it did.

In this paper, I use Marr's levels of analysis to separate two versions of that question. I argue that what our knowledge is about is constrained by the world, so different civilizations facing the same problems are likely to converge on similar content. But how that knowledge is represented is constrained from below, by the human mind and by earlier representations, and is therefore path-dependent. Our representations are not the only possible ones, but they are not arbitrary either, and the ones we inherit decide what is easy to compute next. The *I Ching* offers a contrasting case: a civilization that represented change in a very different way, with consequences that reach from Leibniz's binary arithmetic to everyday reasoning today.

The Truth and Its Representation

Marr (1982) proposed that an information-processing system can be described at three levels: the computational level, which specifies what problem is solved and why; the algorithmic level,

which specifies the representation of the input and output and the steps that transform one into the other; and the implementation level, which describes the physical system that carries out those steps. My high-school question mixed the first two levels. What a theorem says — for example, that for any right triangle, the two shorter sides fix the longest side in the same way — is like the computational level: it specifies what holds, regardless of how anyone writes it. How the theorem is written and proved is like the algorithmic level: a choice of representation and of steps.

The Pythagorean theorem shows this clearly. For every right triangle, the square on the longest side equals the sum of the squares on the other two sides. This was true of every right triangle long before anyone stated it. Yet it has been represented in very different ways. In Euclid's *Elements*, it is proved step by step from axioms and earlier propositions (Euclid, ca. 300 B.C.E./1998). In Chinese mathematics, the same relationship appears as the *gougu* theorem in the *Zhoubi Suanjing*, named after the two sides of a carpenter's square, and a third-century commentary by Zhao Shuang includes a "hypotenuse diagram" that shows the result by arranging four copies of a right triangle inside a square and comparing areas (Cullen, 1996). One representation persuades through a chain of logical steps; the other persuades by letting you see that the pieces fit. The truth is the same. What differs is the path by which a human mind is brought to accept it.

Marr's own example was numbers. The same number can be written in Arabic numerals, Roman numerals, or binary. Thirteen is thirteen in all three, but each representation makes some information explicit and hides other information. Arabic numerals make the decomposition into powers of ten explicit; binary makes powers of two explicit instead. This matters for computation. Multiplying 37 by 24 in Arabic numerals is a routine procedure taught to children. Multiplying XXXVII by XXIV in writing is much harder, because Roman numerals do not line digits up by place value, so the column procedure has nothing to work on. A representation does not change what is true, but it changes what is easy.

In class, we described a representation as a model of the world that guides action. Different models of the same truth can guide action equally well in principle while making some actions much easier in practice. So the answer to "why does the truth appear in this form?" cannot come

from the truth itself. A fact does not choose its own representation. Something else does, and that is where the constraints from below come in.

Constraints from the Mind

The first set of constraints comes from human cognition. We count in base ten, and the most common explanation is anatomical: we have ten fingers, and early counting used them. Other civilizations made other choices. The Babylonians used a base-sixty system, and the Maya used a base-twenty system. These are real examples of the alternative paths I imagined in high school, and one of them still shapes my daily life: an hour has sixty minutes and a circle has 360 degrees because of the Babylonian choice.

Representations also affect the minds that learn them. In Chinese, number names follow the base-ten structure exactly: eleven is 十一 ("ten-one"), twelve is 十二 ("ten-two"), and twenty-one is 二十一 ("two-ten-one"). English hides this structure behind words like "eleven," "twelve," and "twenty." Miller et al. (1995) found that Chinese-speaking preschoolers had an advantage over English-speaking children in learning to count, while the two groups did not differ on other early skills, such as counting small sets. The difference was not in children's mathematical ability but in how transparently their language represented the base-ten system. I learned to count in Chinese and learned English later as a second language, so I never had to discover that "twelve" means "ten and two." My first number words already told me. The representation did part of the computation for me.

Cognitive constraints can also push different civilizations toward similar representations. Place-value notation, in which a digit's value depends on its position, appears to have developed in several places, including Mesopotamia, China, India, and Mesoamerica, and a symbol for zero was developed independently at least twice, by the Maya and in India. If notation were purely arbitrary, this kind of repeated invention would be surprising. It is less surprising if human minds, with limited memory and a need to calculate quickly, face the same pressures everywhere. A system that lets a few symbols express any number, and lets calculation proceed column by column, is simply easier for a human mind to use.

So the mind constrains representation in two directions: it introduces arbitrary features, such as the number ten, and it favours designs, such as place value, that different cultures reach separately.

Constraints from Earlier Representations

The second set of constraints comes from representations that already exist. Once a representation is widely used, later work is built in it, and switching becomes costly; economists call this path dependence (David, 1985). The Babylonian sixty-minute hour has survived for exactly this reason.

Path dependence can also shape what gets discovered. Newton and Leibniz developed calculus independently, with different notations. British mathematicians largely kept Newton's dot notation, while mathematicians on the Continent adopted Leibniz's *dy/dx*, which makes the structure of operations such as the chain rule visible on the page. Historians often link Britain's slower progress in analysis during the eighteenth century partly to this choice. The mathematics was the same on both sides. The representation decided what was easy to do with it, and therefore what got done.

This is the recursive process I sensed in high school. Representations are used to compute; the results of computation are written down as new representations; those become the base for the next round. Tomasello (1999) calls this the ratchet effect: humans keep and build on the modifications made by earlier generations instead of reinventing them, so culture accumulates rather than starting over (see also Tennie et al., 2009). A ratchet only moves forward, which is its strength. But it also means that each new representation inherits the choices built into the old ones. When I learn calculus today, I am also learning, without noticing, the decisions of people who chose its notation three centuries ago.

So the constraints from below are not only the limits of the human brain. They include the whole stack of earlier representations underneath any new idea. Each iteration is free to add something new, but only on top of what is already there. Over many iterations, this narrows the range of paths that remain open, and the process converges toward the civilization we now have.

The *I Ching*: A Representation Built for Change

The *I Ching* (易经) is one of the oldest systems a civilization built to represent the world, and its core ideas read almost like a theory of representation. A Han-dynasty commentary, later elaborated by Zheng Xuan, says that the word *yi* (易) has three meanings: simplicity (简易), change (变易), and constancy (不易). The world is always changing, but the patterns of change are constant, and simple means can capture them.

Those means are two kinds of lines, solid (yang) and broken (yin). The Great Commentary attached to the text says that "one yin and one yang is called the Way" (一阴一阳之谓道), and describes how the system grows: the Great Ultimate produces two modes, the two produce four images, and the four produce eight trigrams (太极生两仪，两仪生四象，四象生八卦; Wilhelm & Baynes, 1967; my translations). Doubling twice more gives sixty-four hexagrams, each with a short text about a kind of situation. In the terms we used in class, this is a symbol system that generates many expressions from a few primitives and a rule. It is partly compositional, too: Tai (泰, "peace") places earth above heaven and Pi (否, "standstill") places heaven above earth, so the same parts in a different arrangement have opposite meanings.

Three further ideas connect the *I Ching* to cognitive science. First, meaning depends on position. Each line is read by its place, from the beginning of a situation at the bottom to its end at the top, which is the same principle as place value in a completely different domain. Second, the commentary makes an explicit claim about the limits of representation: "writing does not fully express speech, and speech does not fully express meaning, so the sages set up images to express meaning fully" (书不尽言，言不尽意……圣人立象以尽意). Like Marr's numerals, this recognizes that different formats make different information available. Third, the basic unit is change itself. A situation is a hexagram that becomes another when one of its lines changes, and "when a thing reaches its limit, it changes; through change it continues" (穷则变，变则通，通则久).

This gives two representations of the same domain. Leibniz's calculus represents change as a continuous rate: *dy/dx* asks how fast one quantity changes with another. The *I Ching* represents change as a qualitative transition between situations: it asks what a situation is turning into, and whether this is the time to act. Neither is false, but each makes a different question feel natural.

One Form, Two Inheritances

Leibniz also links the two traditions directly. He worked out binary arithmetic in 1679, and in 1703 he received from the Jesuit missionary Joachim Bouvet a woodcut of an arrangement of the hexagrams attributed to the eleventh-century philosopher Shao Yong. With broken lines read as 0 and solid lines as 1, the arrangement corresponds to counting from 0 to 63. Leibniz concluded that the ancient Chinese had possessed binary arithmetic and that he had recovered a meaning they had lost (Strickland & Lewis, 2022). Historians generally reject the idea that Leibniz took binary from the *I Ching*, and many doubt that the hexagrams were ever arithmetic. The same form, inherited in China together with texts and questions about timing, was read for meaning; inherited by Leibniz together with place-value arithmetic, it was read as numbers. A representation's computational potential is not contained in its symbols. When Leibniz met the hexagrams, he did not learn a second way of thinking; he translated it into his own.

The *I Ching*'s way of representing change may also have shaped everyday cognition. Peng and Nisbett (1999) describe Chinese dialectical thinking as resting on three principles, that reality is a process of change, that it contains contradictions, and that everything is connected, and they trace these partly to the *I Ching* and to yin and yang. Chinese participants in their studies preferred proverbs containing apparent contradictions and more often resolved conflicts by compromise. Ji et al. (2001) found that Chinese participants expected more change from an initial state than American participants did, and saw people who predicted change as wiser. These are cultural tendencies in judgment, not differences in perception, but they suggest that an inherited representation can shape what ordinary people expect, not only what scholars write.

Would Another Civilization Converge?

Putting these constraints together gives a more precise answer to the question I asked about other civilizations, or other universes. At the level of content, I expect strong convergence. Any civilization that needs to count harvests, predict seasons, or build stable structures faces the same

problems, and the world will push its answers toward the same relationships. The Greek and Chinese traditions reached the same theorem about right triangles, apparently without depending on each other. At the level of representation, I expect much less convergence. The base, the symbols, which concepts are named first, and what counts as a proof all depend on cognitive accidents, such as having ten fingers, and on historical accidents, such as which notation a country's universities happened to adopt.

One objection is that this distinction is too clean. Perhaps some representations are not just convenient but correct. If mathematical truths exist independently of us, as many mathematicians believe, then a good enough representation might be forced on anyone who studies them long enough, and a sufficiently advanced civilization would end up with something very close to ours.

I think this objection is partly right. The repeated invention of place value and zero shows that some representational choices are driven by deep pressures, not accidents. But convergence on design principles is not convergence on form. Babylonian, Mayan, and Indian place-value systems all used position, yet they differed in base, symbols, and how they marked an empty place. Whether or not mathematical truths exist independently, which is a question I cannot settle here, a single truth can still be represented in many ways. What converges is the set of problems a representation must handle, not the representation itself.

A Connected World

This framework also suggests something about the future. Much of my evidence depends on isolation: place value, zero, and the right-triangle theorem were found more than once, which is how we know they reflect deep pressures rather than accidents. The internet and AI are ending that natural experiment. Well-connected groups converge quickly on the same good-enough solutions and lose the diversity needed for further search (Lazer & Friedman, 2007); writers using AI-generated ideas produce stories that are individually more creative but more similar to one another (Doshi & Hauser, 2024); and models trained on AI-generated text lose rare patterns first (Shumailov et al., 2024). In a connected world, agreement becomes weaker evidence that an answer was forced, and, as Leibniz's reading of the hexagrams already showed, contact can turn two traditions into one before anyone notices what was lost.

Why We Have to Learn This

The other part of my high-school question was more personal: why do we have to learn things we are not interested in? This analysis gives an answer I did not have at the time. Learning a representation is a way of inheriting computation. When I learned place value, algebraic notation, or calculus, I was not just memorizing arbitrary rules. I was receiving tools that many earlier people had built so that I would not have to solve their problems again. That is the ratchet working as it should.

But it also explains why it rarely felt that way. Representations arrive in school without the problems they were created to solve. A student sees *dy/dx* but not the problems that made Leibniz's notation worth fighting over. Without the problem, a representation looks like an arbitrary rule handed down for no reason, and "why do we have to learn this?" is a reasonable question. In Marr's terms, school often teaches the algorithmic level without the computational level: the steps, without the question they answer.

Looking back, my high-school question was less a trap than a mix-up of levels. The form in which a truth appears does not come from the truth alone, but from the minds that use it and the representations that came before. Our knowledge could have been different, in form much more than in content. And I had to learn it because it was the fastest way to stand on the work of everyone before me, even if no one showed me what that work had been for.

If another civilization had started differently, its students would probably be learning different symbols. I suspect they would still be asking the same question.

References

Cullen, C. (1996). *Astronomy and mathematics in ancient China: The Zhou bi suan jing*. Cambridge University Press.

David, P. A. (1985). Clio and the economics of QWERTY. *The American Economic Review, 75*(2), 332–337.

Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances, 10*(28), Article eadn5290. https://doi.org/10.1126/sciadv.adn5290

Euclid. (1998). *Euclid's Elements* (D. E. Joyce, Ed.). Clark University. https://mathcs.clarku.edu/~djoyce/elements/elements.html (Original work published ca. 300 B.C.E.)

Ji, L.-J., Nisbett, R. E., & Su, Y. (2001). Culture, change, and prediction. *Psychological Science, 12*(6), 450–456. https://doi.org/10.1111/1467-9280.00384

Lazer, D., & Friedman, A. (2007). The network structure of exploration and exploitation. *Administrative Science Quarterly, 52*(4), 667–694. https://doi.org/10.2189/asqu.52.4.667

Marr, D. (1982). *Vision: A computational investigation into the human representation and processing of visual information*. W. H. Freeman.

Miller, K. F., Smith, C. M., Zhu, J., & Zhang, H. (1995). Preschool origins of cross-national differences in mathematical competence: The role of number-naming systems. *Psychological Science, 6*(1), 56–60. https://doi.org/10.1111/j.1467-9280.1995.tb00305.x

Peng, K., & Nisbett, R. E. (1999). Culture, dialectics, and reasoning about contradiction. *American Psychologist, 54*(9), 741–754. https://doi.org/10.1037/0003-066X.54.9.741

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature, 631*(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Strickland, L., & Lewis, H. R. (2022). *Leibniz on binary: The invention of computer arithmetic*. MIT Press.

Tennie, C., Call, J., & Tomasello, M. (2009). Ratcheting up the ratchet: On the evolution of cumulative culture. *Philosophical Transactions of the Royal Society B: Biological Sciences, 364*(1528), 2405–2415. https://doi.org/10.1098/rstb.2009.0052

Tomasello, M. (1999). *The cultural origins of human cognition*. Harvard University Press.

Wilhelm, R., & Baynes, C. F. (Trans.). (1967). *The I Ching or book of changes* (3rd ed.). Princeton University Press.